

# Sales Forecasting & Demand Prediction

Final Report

# Members

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# Contents

01 Executive Summary

02 Project Overview

03 Business Impact

04 Data Overview & Preprocessing

- **Summary**
- **Data Loading**
- **Cleaning Steps**

05 Exploratory Data Analysis (EDA)

- **Critical Business Insights**
- **Visualization Methodology**

06 Advanced Data Analysis and Feature Engineering

- **Summary**
- **Advanced Data Analysis**
- **Feature Engineering**
- **Data Visualization**
- **Result & Insights**

07 Modeling

- **Model Improvement**
- **Key Insights**
- **Result**
- **Model Improvement**

08 Deployment

- **MLOps Implementation**
- **Model Deployment**
- **Model Monitoring and Retraining Strategy**

# Executive Summary

This project aims to build a predictive system for forecasting sales and demand using historical transaction and product data.

The objective is to help businesses reduce stockouts, optimize inventory levels, and improve operational planning.

The project includes data cleaning, exploratory data analysis (EDA), feature engineering, statistical testing, and a modeling plan using machine learning and time-series approaches.

Final results indicate that the engineered features and cleaned dataset are suitable for accurate forecasting, with recommended next steps for model deployment and business integration.

# Project Overview

The Sales Forecasting and Demand Prediction project aims to build a machine learning model that predicts future sales and demand for products based on historical data. Accurate forecasting helps businesses optimize inventory management, staffing, and marketing strategies. This project will apply data science techniques, from data collection and analysis to model deployment and monitoring, enabling businesses to make data-driven decisions.

## Business Impact

This project applies data science techniques from data collection and analysis to model deployment and monitoring, enabling businesses to make data-driven decisions that directly impact their bottom line. By implementing advanced forecasting models, businesses can reduce inventory costs, improve customer satisfaction through better stock availability, optimize workforce allocation, and enhance marketing campaign effectiveness.

# DATA OVERVIEW & PREPROCESSING

Milestone 1 establishes the foundation of the Sales Forecasting and Demand Prediction project. The work includes data collection, cleaning, exploratory data analysis (EDA), feature engineering, and preparing the dataset for building machine learning models in the next stages.

The dataset consists of historical sales transactions containing details about products, customers, dates, locations, and key business metrics such as sales, quantity, discount, and profit.

## Data Loading:

- The dataset was loaded from a CSV file (Original Data.csv). Initial inspection confirmed it contains 51,931 records and 24 columns.
- Original Size: 51,931 rows, 24 columns
- Final Cleaned Dataset: 51,290 rows, 31 columns

## Cleaning Steps:

- Dropped non-essential columns (Row ID, Order ID, etc.)
- Removed duplicates and rows with excessive nulls
- Imputed missing values for Profit, Order Priority, and Market
- Managed outliers in Sales, Discount, Profit, and Shipping Cost using median/mean replacement

# EXPLORATORY DATA ANALYSIS (EDA)

Perform exploratory data analysis (EDA) to understand sales trends, seasonality, and external factors.

Create visualizations (line plots, bar charts, heatmaps) to identify trends and patterns.

## Critical Business Insights

- Strong inverse relationship between discount levels and profit margins
- Furniture category demonstrates revenue strength but profit weakness
- Technology products lead in profitability across all categories
- Consumer segment accounts for majority of market share
- Sustained sales growth trajectory across four-year period
- Seasonal performance peaks during autumn months
- Consistent sales performance during weekends and holidays

## Visualization Methodology

- Statistical distribution analysis of key financial metrics
- Comparative performance visualization across product categories
- Geographic performance mapping across regions and markets
- Temporal trend analysis with seasonal decomposition
- Correlation matrix analysis for variable relationships
- Customer segment contribution analysis through proportional visualization
- Anomaly detection and treatment in critical financial indicators

# Advanced Data Analysis and Feature Engineering

This milestone focused on deepening the analysis and creating advanced features to improve forecasting accuracy.

Building on the cleaned dataset from Milestone 1, the goal was to extract trends, seasonality, and external influences.

## Advanced Data Analysis

- **Time Series Analysis:** Monthly and weekly sales showed clear seasonal peaks and an overall upward trend.
- **Stationarity Testing:** ADF test applied to ensure data suitability for ARIMA and other time-series models.
- **Correlation Analysis:** Promotional activities significantly impacted sales but sometimes reduced profit margins. Certain product categories correlated with seasonal patterns.

## Feature Engineering

- **Lag & Rolling Features:** Previous day, 7-day, and 30-day sales; rolling mean and standard deviation.
- **Seasonal & Temporal Features:** Month, quarter, season, holidays, and weekends.
- **External Factors & Promotions:** Flags for discounts and special events.
- **Transformations:** Scaled numerical features; one-hot and target encoding for categorical features; aggregated monthly sales per category.



# Advanced Data Analysis and Feature Engineering

## Data Visualization

- **Trends:** Line charts show long-term growth and seasonal peaks.
- **External Factors:** Promotions increase sales; adverse weather reduces demand.
- **Dashboards:** Interactive tools for exploring time, category, region, and promotions.

## Result & Insights

- Advanced analysis and feature engineering improved the forecasting model's accuracy.
- Lag features, rolling means, and external variables captured short-term fluctuations and seasonal patterns.
- Correlation and visualization analyses highlighted key sales drivers such as promotions, seasons, and regions.
- These insights support accurate, business-relevant forecasting and decision-making.

# Modeling

## Model Improvements

- Enhanced model accuracy through better feature engineering (lags, rolling averages, seasonal features).
- Added holiday and time-based indicators to capture recurring patterns.
- Incorporated external factors like promotions and events to improve prediction stability.
- Improved data preprocessing reduced noise and strengthened model performance.

## Key Insights

- Strong correlations found between promotions and higher sales.
- Seasonal patterns (holidays, month trends) significantly influenced demand.
- Visual analysis revealed consistent peaks and dips tied to specific time periods.
- External conditions occasionally shifted demand, showing their value as model inputs.

## Results and Insights

- The improved models delivered more stable and accurate forecasts.
- Better detection of short-term changes and long-term trends.
- Insights support stronger planning in inventory, demand forecasting, and promotions.

# Modeling

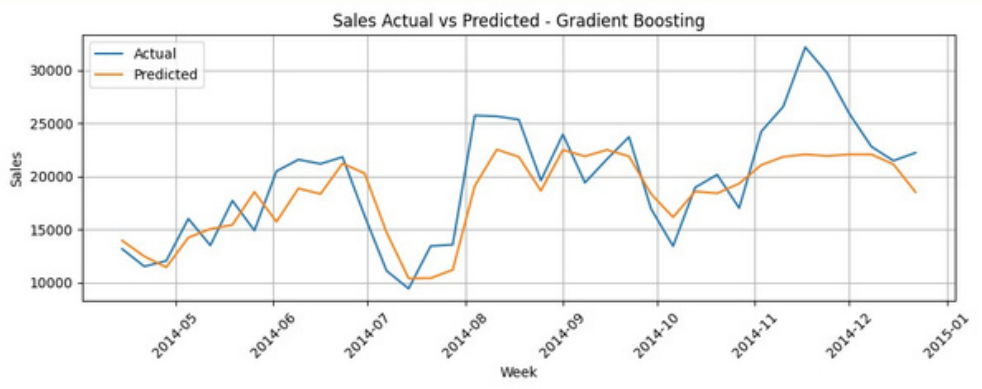
## Model Performance

The tables below summarize the performance of the different forecasting models evaluated on the dataset.

The metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE),  $R^2$  Score, and Mean Absolute Percentage Error (MAPE), which collectively provide insights into prediction accuracy and reliability.

### Sales Model :

|                   | MAE         | RMSE        | R2        | MAPE     |
|-------------------|-------------|-------------|-----------|----------|
| Gradient Boosting | 2708.535124 | 3408.931467 | 0.609793  | 0.135969 |
| ETS               | 2885.938566 | 3645.230715 | 0.553821  | 0.152431 |
| XGBoost           | 2879.008608 | 3712.141307 | 0.537291  | 0.138989 |
| Naive             | 3235.644624 | 4070.598037 | 0.443615  | 0.174023 |
| Random Forest     | 3420.907401 | 4342.636989 | 0.366764  | 0.163943 |
| Decision Tree     | 4064.430408 | 4919.259346 | 0.187435  | 0.211713 |
| ARIMA(1,1,1)      | 6467.012820 | 7898.722914 | -1.094944 | 0.296525 |

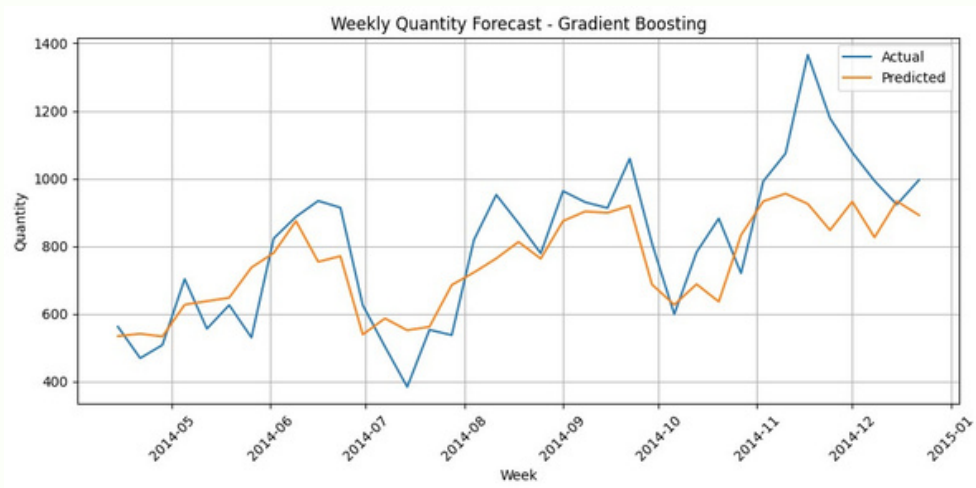


# Modeling

## Model Performance

Quantity Model :

|                   | MAE         | RMSE        | R2        | MAPE     |
|-------------------|-------------|-------------|-----------|----------|
| XGBoost           | 1876.059364 | 2609.215161 | 0.587627  | 0.127318 |
| Gradient Boosting | 1890.948210 | 2644.144384 | 0.576513  | 0.130679 |
| Random Forest     | 1895.034298 | 2650.860288 | 0.574359  | 0.126579 |
| ETS               | 2105.764252 | 2689.473107 | 0.561868  | 0.148172 |
| Decision Tree     | 2171.211392 | 3192.308931 | 0.382723  | 0.163658 |
| Naive             | 3096.132674 | 3608.044668 | 0.211478  | 0.209276 |
| ARIMA(1,1,1)      | 6239.788279 | 7417.304832 | -2.332441 | 0.372415 |



# Deployment

*Implement MLOps practices and deploy the forecasting model for real-time or batch predictions*

## MLOps Implementation

- The project integrates MLflow to manage experiments, track model versions, and ensure reproducibility. Model training and evaluation metrics—including MAE, MSE, RMSE, and  $R^2$ —are logged for traceability.
- Key artifacts, such as plots comparing actual vs. predicted sales, are saved and tracked to support transparency. The final model, along with its input-output signature, is registered in MLflow, enabling version control and structured deployment for future use.

## Model Deployment

- The final sales forecasting model has been deployed as a FastAPI web service, providing endpoints for predicting sales and demand. Users can send structured inputs to the API and receive real-time sales forecasts.
- An interactive Streamlit dashboard has been built to allow business users to easily input relevant sales and contextual data (e.g., quantity, discount, category, season, region) and view predicted sales results. The dashboard communicates with the FastAPI service to fetch predictions, offering a user-friendly interface for real-time forecasting and decision-making.

# Deployment

Implement MLOps practices and deploy the forecasting model for real-time or batch predictions

## Model Monitoring and Retraining Strategy

Machine learning models in production can lose accuracy over time due to changes in data, customer behavior, seasons, or external factors. Model monitoring continuously tracks the model's performance, stability, and data quality to ensure predictions remain accurate and reliable.

Key issues addressed include:

- Performance degradation: accuracy declines as the model encounters new patterns.
- Data drift: changes in the input data distribution affect model predictions.
- Concept drift: evolving relationships between features and the target over time.

The monitoring workflow involves collecting incoming data, generating predictions, storing results, comparing with actual outcomes, calculating performance metrics, detecting anomalies, and triggering alerts. Combined with a retraining strategy, this ensures long-term accuracy, stability, and business value.